

Ziteng Song

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Education

Cornell University

B.A. in Computer Science; Cumulative GPA: 4.00

Teaching Assistant: CS 4620 — Introduction to Computer Graphics

Teaching Assistant: CS 4621 — Computer Graphics Practicum

Expected May 2027

Ithaca, NY

Fall 2026

Fall 2026

Technical Skills

Programming Languages: Python, C++, Java, TypeScript, GLSL

Machine Learning & Data: PyTorch, NumPy, Pandas, SciPy; **Techniques:** Time-Series Forecasting, Feature Engineering, PCA

Computer Vision & 3D: OpenCV, COLMAP, gsplat; **Techniques:** Optical Flow, Multi-View Geometry, 3D Gaussian Splatting

Software Engineering: Git, Docker

Graphics & Game Development: WebGL 2, Three.js; **Techniques:** GPU Simulation, Path Tracing, Heterogeneous Volumetric Rendering

Experience

Cornell University – Imaging and HCI Group

Jan 2026 – Present

Undergraduate Researcher, Advisor: Prof. Abe Davis – Dynamic 3D Gaussian Splatting

Ithaca, NY

- Developed a PyTorch pipeline with gsplat to recover 3D vibration modes from unsynchronized multi-view videos through 3D Gaussian Splatting, enabling dynamic scene reconstruction and interactive motion synthesis.
- Enabled joint reconstruction from unsynchronized multi-view video inputs through frequency-domain motion analysis.
- Designed a graph neural network with learnable control-point embeddings to predict spatially coherent 3D displacement fields for foreground Gaussians.
- Reconstructed vibrating scenes from 720p videos in ~1 hour on an RTX 5090, versus ModalNeRF's reported ~5 hours on an A6000, and outperformed PhysGaussian, PhysDreamer, and ModalNeRF in motion reconstruction accuracy and physical plausibility of novel-motion synthesis.

Wanka Huanju Culture Media Co., Ltd (Beijing)

May 2025 – Aug 2025

Machine Learning Intern

Beijing, China

- Engineered an auditable preprocessing pipeline for 562K advertising records across 5.6K product-placement time series, correcting aggregation errors and implementing calendar alignment, missingness masks, and active-period segmentation.
- Fine-tuned the Moirai time-series foundation model with LoRA in PyTorch, updating only 0.53% of parameters for 7-day forecasting from 90-day histories; achieved 23.20% WAPE in historical backtesting, reducing WAPE by 3.7% relative to a 7-day-mean baseline and 2.9% relative to zero-shot Moirai, averaged across three training seeds.
- Contributed to an internal knowledge base with Q&A functionality, integrating Dify and VikingDB with a WeChat Official Account; by May 2026, 9 employee cohorts had used it to prepare for and pass recurring internal assessments.

Projects

AquaFlow: WebGL-Based Fluid Simulation & Rendering | TypeScript, GLSL, WebGL 2, Three.js

Nov 2025 – Dec 2025

- Developed a real-time SPH-style fluid simulation and rendering on a two-person team using WebGL 2 and the course-provided AniGraph framework, achieving approximately 120 FPS with 40,000 particles on an NVIDIA RTX 4070 Laptop GPU.
- Developed a GPU data layout that stored particle positions, velocities, and densities in floating-point textures and updated them through shader passes with ping-pong buffering, avoiding per-frame CPU readback and re-upload.
- In an optimized physics-only configuration with spatial hashing and bitonic sorting, the system ran 20,000 particles at approximately 90 FPS on the M2 MacBook Air, versus 4,000 particles at 30 FPS in a CPU-based configuration. Resolved stale-density instability by inserting an explicit density pass and bounding physics substeps.
- Rendered water texture using volumetric ray-marching, Beer-Lambert attenuation, isosurface detection, toon-style Blinn-Phong highlights, and SDF-based shoreline foam.

Gnome More Shrooms: Mobile Tower Defense Game | C++

Jan 2026 – May 2026

- Co-led programming and core gameplay design for a mobile tower-defense game on a 10-person team; the game earned Runner-Up (2nd place) in Game of the Year—Mobile at the 2026 Rookie Awards.
- Built interconnected gameplay systems spanning crop-based resource production, movable auto-attacking turrets, enemy wave spawning, map expansion, and base defense.
- Introduced runtime profiling with per-stage timings, draw-call and cache diagnostics, and p95/p99 frame-time tracking; improved cache reuse during camera movement, visibility culling, and pause/resume timing, largely eliminating sudden drops below 60 FPS during gameplay testing.

Physically Based C++ Path Tracer | C++, Path Tracing, Volumetric Rendering

Jan 2026 – May 2026

- Extended the Nori educational framework into a physically based C++ renderer, implementing BVH acceleration, light and BSDF sampling with MIS, path tracing, dielectric and microfacet materials, texture mapping, and environment lighting.
- Implemented heterogeneous participating media with delta tracking and Henyey-Greenstein phase functions; rendered a 4000×3000 final scene at 8,192 spp.
- Passed three chi-square tests for volumetric sampling at $\alpha = 0.01$ (up to 200K samples per test); evaluated 10 image cases spanning five Mitsuba comparisons, one analytic reference, and four controlled procedural-cloud variants.